



ARBA MINCH UNIVERSITY

ARBA MINCH INSTITUTE OF TECHNOLOGY
CIVIL ENGINEERING FACULTY

Worksheet: -Contract, Specification and Quantity Surveying

Course instructor–Mr. Kinfu M.

Date of Submission – 22/03/ 2024

Total mark = 15%

Target Group: G4-(CEng)

Credit hour - 2-3-0.

Semester - I - 2023/4

Instruction:

☛ *Mandatory to use standard formats.*

Part-I: Give short and brief answer (5 marks each)

- 1) Discuss different types of contract.
- 2) Discuss different types of project delivery system with respective advantage and disadvantage
- 3) Discuss qualities of good specification, and mention principles or guidelines while writing specification.
- 4) Mention information, knowledge, managerial skills required for construction cost estimation.
- 5) What is procurement? Discuss types of procurement.

Part-II: (Take-off sheet and Bar schedule) (16 Marks)

- 6) Prepare take-off /bar schedule for work items listed below for the given drawings of some part or section of a building? **(4 marks each)**
 - A. BOQ for the substructural works for the given drawing A) Floor plan only.
 - B. C-30 Concrete for the drawings :- B) The Footing and C) The Stair Section
 - C. Formwork for the drawing:- B) The Footing and C) The Stair Section
 - D. Reinforcement bar for the drawing:- B) The Footing

Part-III: (Rate Analysis)..... (9 Marks)

- 7) Suppose you are in tendering process. Determine the Unit Rate for C-20 concrete (1:3:4) for ground slab using the information given below. The water to cement ration for the mix is 0.60. The productivity of the Labor and Equipment crew assigned for the work is estimated to 15.0 m³/day. From market information Cement is purchased at 1350 birr per quintal including loading cost and unloaded by 10 birr per quintal. A Sino (16m³) of sand is purchased at 12,000 birr. Coarse aggregate is purchased at 600 birr per m³ at the crusher site and costs 2700 birr for both loading and

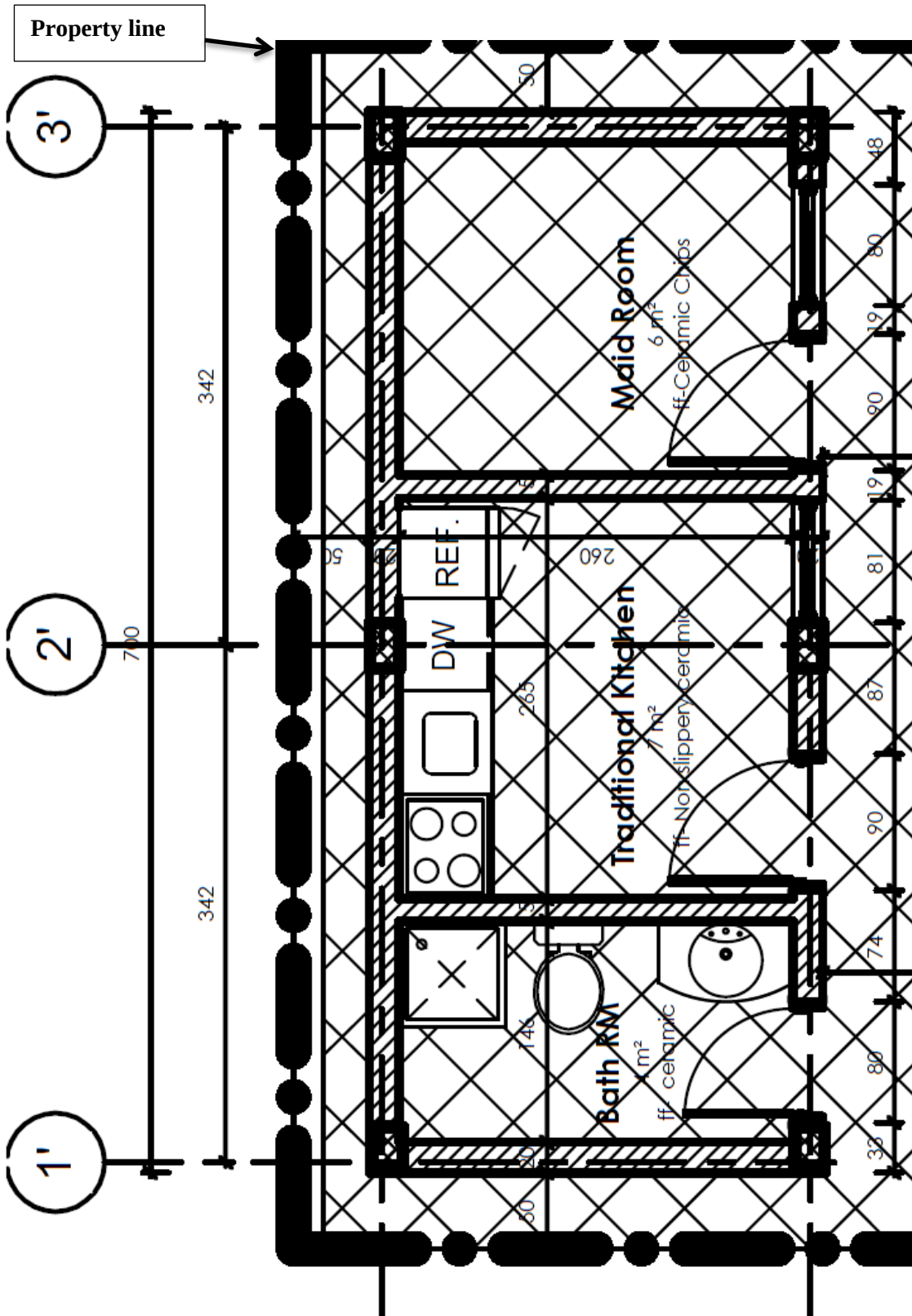
transportation of 1 **Sino** of it. Take water supplies rate of 7 birr per m³ and normal working hours a day. Consider 10% over-head, 15% Profit allowance, and 40% for Wastage and shrinkage.

Table-1: Labor Data				
Labor by trade	No.	UF	Hourly Labor Wage	Index factor
Forman	1	0.5	60 Birr/hr	1.5
Mason	1	0.5	50 Birr/hr	1.5
Daily Labor	8	1.00	30 Birr/hr	1.3
Mixer Operator	1	0.5	30 Birr/hr	1.3

Table-2: Equipment Data				
Equipment Type	No.	UF	Equipment Owning cost	Equipment Operation cost
Mixer	1	0.5	250 Birr/hr	25 Birr/hr
Vibrator	1	0.5	50 Birr/hr	5 Birr/hr

-----**GOOD LUCK**-----

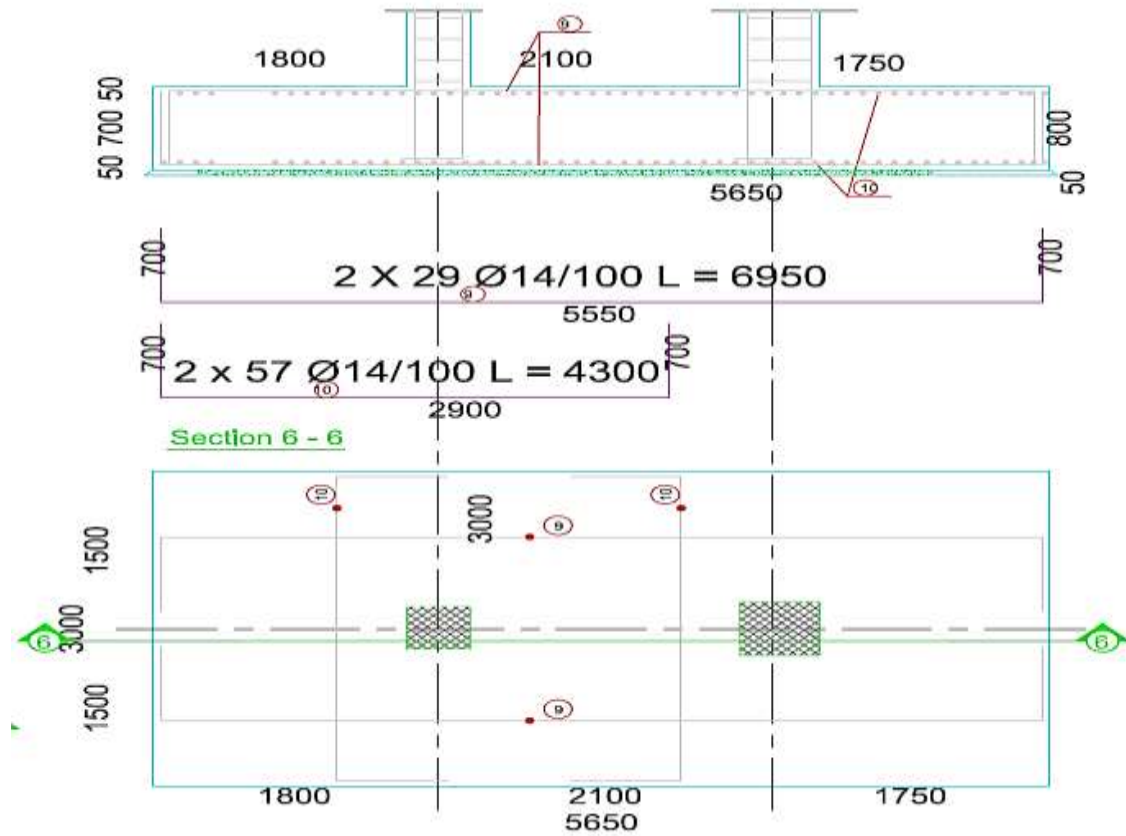
A) Floor Plan



Given

1. Backfill under masonry= 20cm
2. Hard core= 20cm
3. Masonry width= 50cm
4. Masonry depth= 50cm
Below NGL & 40cm above NGL
5. Grade beam section =200 X 300
6. Slab depth= 10cm

B) Footing



C) Stair Section

